

# POWERLINE ULTRA (PUD DOUBLE SKIN)

In-Line Mixed-Flow Fan

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**ELTA**



# POWERLINE ULTRA (PUD DOUBLE SKIN)

## Description

The PowerLine Ultra (PUD double skin) of In-Line Mixed-Flow Fans is designed for a wide range of duct mounted applications. They are most suitable in commercial and industrial applications where medium pressure is required. They are available in various speed options and 7 sizes, extending from 315 to 630mm diameter.

### Typical Applications

Commercial and industrial supply or exhaust air applications such as shopping centres, office buildings, exhibition centres, hotels, health centres, schools and universities.

### Features

- Corrosion resistant aluminium frame with fibreglass reinforced corners for strength and durability.
- Unit is constructed from double insulated panels filled with 25mm polyurethane (pu) insulation reducing radiated noise.
- Removable side panel, provides easy access for onsite inspection and maintenance.
- standard direct-drive TEFC motors.
- Large choice of speeds available.
- To improve energy efficiency, motors can be speed-controlled.
- Motors complying with Ex d, Ex e, Ex nA and Ex tD Standards can be fitted.
- A range of matched ancillaries is also available.

### Construction

The casing shall be constructed from self-supporting corrosion-resistant extruded aluminium section assembled with fibreglass reinforced corners and 25mm thick double skinned panels. Double skin panels are manufactured from pre-painted galvanised steel for outer skin and pre-galvanised steel inner skin, in-filled with polyurethane (PU) injected insulation.

Mixed-flow impellers are made from high performance injection moulded composite material.

### Motors

- Supplied with standard direct-drive TEFC motors
- Electricity supply - single or three-phase to suit a wide range of voltages and frequencies
- Ball Bearing - sealed for life
- Standard direct-drive TEFC motors can be single or multi-speed and can be speed-controlled using a variable speed drive

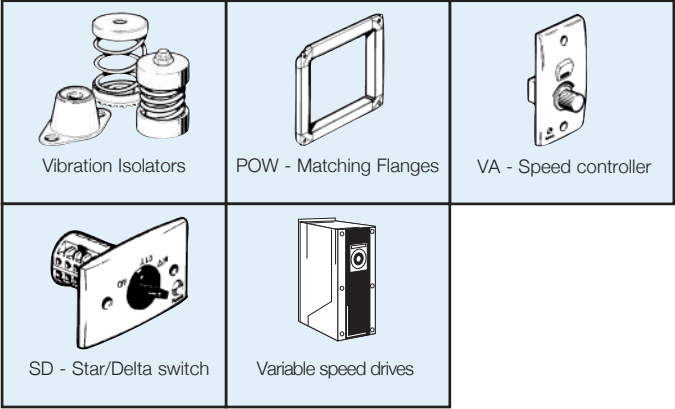
### Internal Thermal Protection

- Optional extra on standard frame motors

### Testing

- Airflow tests to ISO5801:2007
  - Noise tests to BS848:Part 2, 1985
- \*Performance tests & data are based on PowerLine Ultra

## Ancillary Equipment



## Suggestion Specification

The duct mounted fans shall be of the In-line Mixed-Flow PowerLine Ultra (PUD double skin) as designed and manufactured by Elta and be of the model numbers shown on the schedule/drawings.

They shall include double skin cabinet housing with 25mm PU injected panels.

The Mixed-Flow impellers shall be made from high performance injected moulded composite plastic and driven by standard direct-drive motors as nominated.

All performance data shall be for a complete assembled unit based on ISO5801 for air flow and BS848: Part 2, 1985 or ISO13347-3 for noise.

\*Performance tests & data are based on PowerLine Ultra

## How To Order

|                                             |               |             |             |             |             |             |               |
|---------------------------------------------|---------------|-------------|-------------|-------------|-------------|-------------|---------------|
|                                             | <div>PU</div> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | <div>DS</div> |
| PowerLine Ultra                             |               |             |             |             |             |             |               |
| D - 3 Phase                                 |               |             |             |             |             |             |               |
| E - 1 Phase                                 |               |             |             |             |             |             |               |
| Fan Diameter (cm)                           |               |             |             |             |             |             |               |
| Fan Speed, no. of poles                     |               |             |             |             |             |             |               |
| DD - Standard Motor                         |               |             |             |             |             |             |               |
| Special Requirements e.g. epoxy finish etc. |               |             |             |             |             |             |               |
| Double Skin                                 |               |             |             |             |             |             |               |

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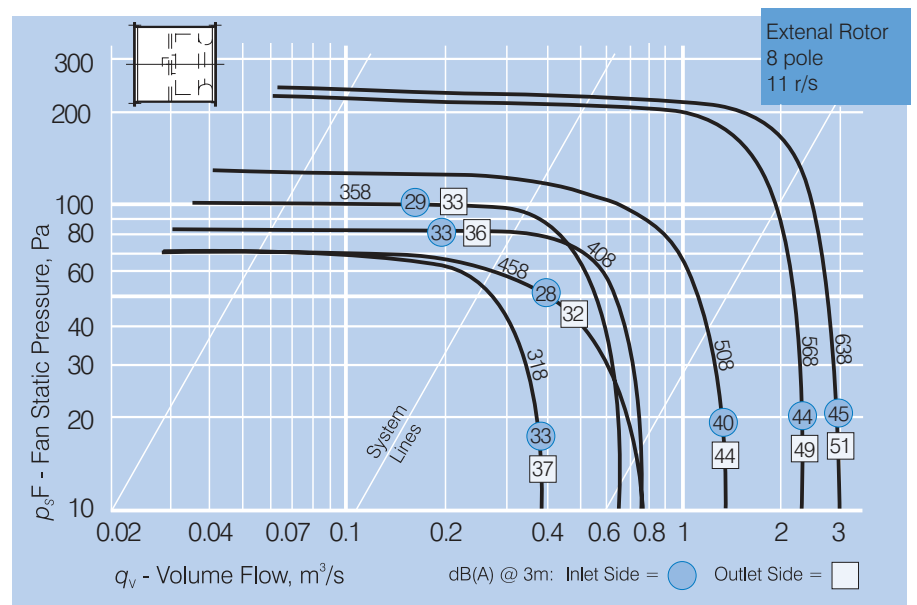
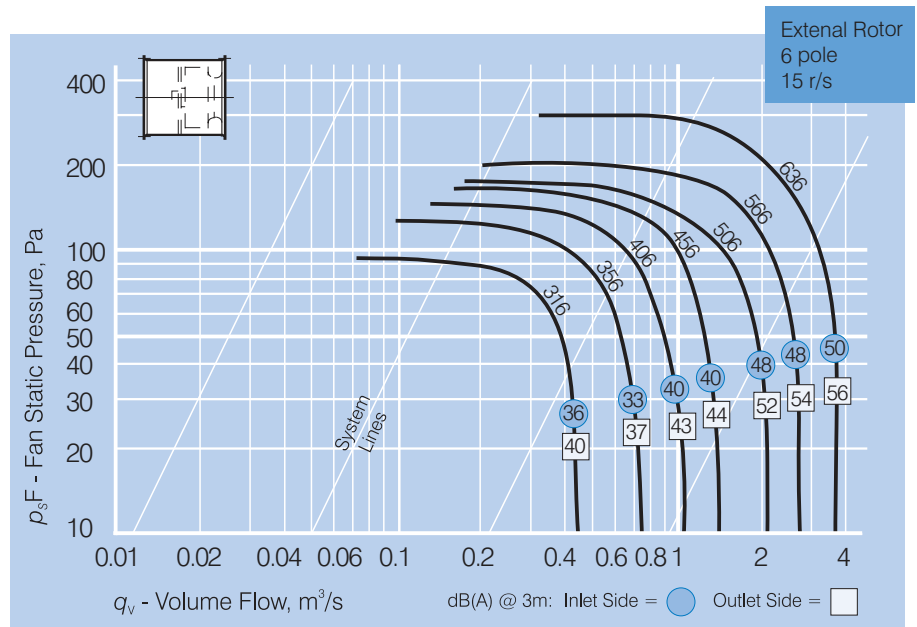
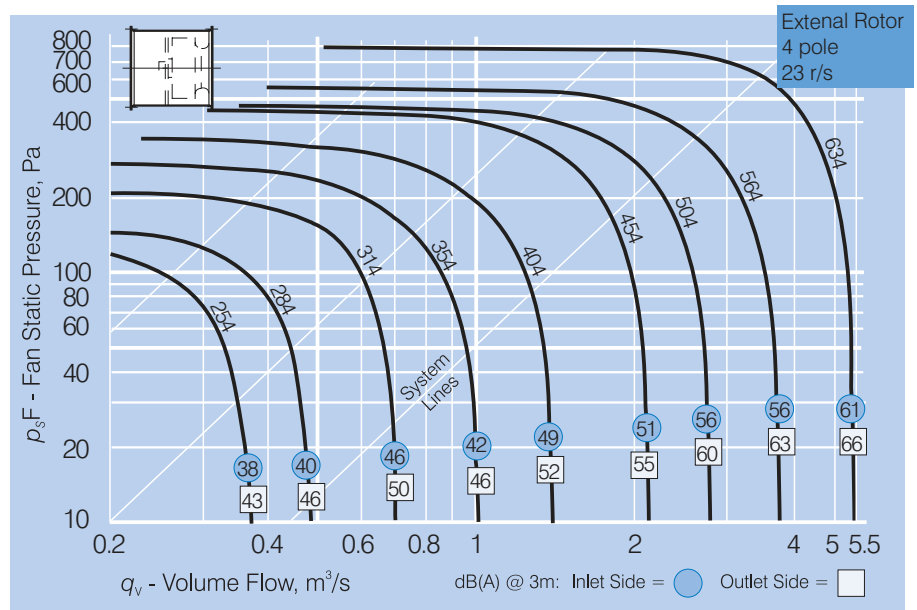
## Note

Performance curves shown are based on nominal speeds.

As motor speeds may vary from one manufacturer to another, and from one motor type to another, it is possible actual fan speeds may differ and, therefore, the performance of the fan.

## Dimensions

| Model       | Dimensions, mm |     |     |     | App. wt. kg* |
|-------------|----------------|-----|-----|-----|--------------|
|             | A              | B   | C   | D   |              |
| PUD...DD-DS |                |     |     |     |              |
| PUE...DD-DS |                |     |     |     |              |
| 31          | 400            | 460 | 460 | 520 | 32           |
| 35          | 450            | 510 | 510 | 570 | 36           |
| 40          | 500            | 510 | 560 | 570 | 48           |
| 45          | 550            | 560 | 610 | 620 | 55           |
| 50          | 650            | 660 | 710 | 720 | 67           |
| 56          | 725            | 760 | 785 | 820 | 101          |
| 63          | 800            | 810 | 860 | 870 | 138          |



# POWERLINE ULTRA (PUD DOUBLE SKIN)

## Technical Data & Noise Levels

| Model<br>PUD...DD-DS<br>PUE...DD-DS | Nom.<br>speed,<br>rev/sec |        | PUD/PUE...<br>DD-DS<br>AVG. dB(A)<br>@ 3m | PUE...DD-DS<br>1 ph. |      | PUD...<br>DD-DS<br>3 ph.<br>kW* | In-Duct Sound Power Levels, dB |     |     |     |    |    |    |    |
|-------------------------------------|---------------------------|--------|-------------------------------------------|----------------------|------|---------------------------------|--------------------------------|-----|-----|-----|----|----|----|----|
|                                     |                           |        |                                           | kW                   | Amps |                                 | 63                             | 125 | 250 | 500 | 1k | 2k | 4k | 8k |
| 314                                 | 23                        | Inlet  | 46                                        | 0.25                 | 1.80 | 0.37                            | 74                             | 70  | 64  | 62  | 60 | 60 | 59 | 53 |
|                                     |                           | Outlet | 50                                        |                      |      |                                 | 74                             | 71  | 64  | 65  | 64 | 66 | 61 | 56 |
| 316                                 | 15                        | Inlet  | 36                                        | 0.12                 | 1.05 | 0.37                            | 61                             | 60  | 54  | 52  | 50 | 50 | 49 | 43 |
|                                     |                           | Outlet | 40                                        |                      |      |                                 | 64                             | 61  | 54  | 55  | 54 | 56 | 51 | 46 |
| 318                                 | 13                        | Inlet  | 33                                        | -                    | -    | -                               | 58                             | 57  | 51  | 49  | 47 | 47 | 46 | 40 |
|                                     |                           | Outlet | 37                                        |                      |      |                                 | 61                             | 58  | 51  | 52  | 51 | 53 | 48 | 43 |
| 354                                 | 23                        | Inlet  | 42                                        | 0.25                 | 1.80 | 0.37                            | 65                             | 68  | 61  | 58  | 54 | 56 | 55 | 49 |
|                                     |                           | Outlet | 46                                        |                      |      |                                 | 66                             | 67  | 61  | 60  | 62 | 60 | 58 | 52 |
| 356                                 | 16                        | Inlet  | 33                                        | 0.12                 | 1.05 | 0.37                            | 56                             | 59  | 52  | 49  | 45 | 47 | 46 | 40 |
|                                     |                           | Outlet | 37                                        |                      |      |                                 | 57                             | 58  | 52  | 51  | 53 | 51 | 49 | 43 |
| 358                                 | 13                        | Inlet  | 29                                        | -                    | -    | -                               | 52                             | 55  | 48  | 45  | 41 | 43 | 42 | 36 |
|                                     |                           | Outlet | 33                                        |                      |      |                                 | 53                             | 54  | 48  | 47  | 49 | 47 | 45 | 39 |
| 404                                 | 22                        | Inlet  | 49                                        | 0.75                 | 4.90 | 0.75                            | 69                             | 73  | 66  | 66  | 60 | 63 | 62 | 56 |
|                                     |                           | Outlet | 52                                        |                      |      |                                 | 71                             | 71  | 66  | 67  | 68 | 67 | 64 | 58 |
| 406                                 | 15                        | Inlet  | 40                                        | 0.18                 | 1.55 | 0.37                            | 60                             | 64  | 57  | 57  | 51 | 54 | 53 | 47 |
|                                     |                           | Outlet | 43                                        |                      |      |                                 | 62                             | 62  | 57  | 58  | 59 | 58 | 55 | 49 |
| 408                                 | 11                        | Inlet  | 33                                        | -                    | -    | -                               | 53                             | 57  | 50  | 50  | 44 | 47 | 46 | 40 |
|                                     |                           | Outlet | 36                                        |                      |      |                                 | 55                             | 55  | 50  | 51  | 52 | 51 | 48 | 42 |
| 454                                 | 23                        | Inlet  | 51                                        | -                    | -    | 1.10                            | 73                             | 75  | 70  | 67  | 64 | 65 | 63 | 58 |
|                                     |                           | Outlet | 55                                        |                      |      |                                 | 78                             | 74  | 72  | 71  | 71 | 68 | 65 | 59 |
| 456                                 | 15                        | Inlet  | 40                                        | -                    | -    | 0.37                            | 62                             | 64  | 59  | 56  | 53 | 54 | 52 | 47 |
|                                     |                           | Outlet | 44                                        |                      |      |                                 | 67                             | 63  | 61  | 60  | 60 | 57 | 54 | 48 |
| 458                                 | 9                         | Inlet  | 28                                        | -                    | -    | -                               | 50                             | 52  | 47  | 44  | 41 | 42 | 40 | 35 |
|                                     |                           | Outlet | 32                                        |                      |      |                                 | 55                             | 51  | 49  | 48  | 48 | 45 | 42 | 36 |
| 504                                 | 21                        | Inlet  | 56                                        | -                    | -    | 1.50                            | 77                             | 81  | 76  | 71  | 70 | 69 | 66 | 61 |
|                                     |                           | Outlet | 60                                        |                      |      |                                 | 85                             | 81  | 82  | 73  | 76 | 72 | 67 | 60 |
| 506                                 | 14                        | Inlet  | 48                                        | -                    | -    | 0.55                            | 69                             | 73  | 68  | 63  | 62 | 61 | 58 | 53 |
|                                     |                           | Outlet | 52                                        |                      |      |                                 | 77                             | 73  | 74  | 65  | 68 | 64 | 59 | 52 |
| 508                                 | 11                        | Inlet  | 40                                        | -                    | -    | -                               | 61                             | 65  | 60  | 55  | 54 | 53 | 50 | 45 |
|                                     |                           | Outlet | 44                                        |                      |      |                                 | 69                             | 65  | 66  | 57  | 60 | 56 | 51 | 44 |
| 564                                 | 22                        | Inlet  | 56                                        | -                    | -    | 3.00                            | 79                             | 81  | 77  | 72  | 71 | 71 | 69 | 64 |
|                                     |                           | Outlet | 63                                        |                      |      |                                 | 82                             | 84  | 80  | 79  | 78 | 77 | 75 | 69 |
| 566                                 | 15                        | Inlet  | 48                                        | -                    | -    | 0.75                            | 73                             | 73  | 69  | 64  | 63 | 64 | 59 | 52 |
|                                     |                           | Outlet | 54                                        |                      |      |                                 | 75                             | 76  | 71  | 70  | 70 | 69 | 64 | 57 |
| 568                                 | 13                        | Inlet  | 44                                        | -                    | -    | -                               | 69                             | 73  | 67  | 61  | 59 | 57 | 53 | 46 |
|                                     |                           | Outlet | 49                                        |                      |      |                                 | 72                             | 73  | 68  | 67  | 66 | 63 | 58 | 50 |
| 634                                 | 24                        | Inlet  | 61                                        | -                    | -    | 5.50                            | 71                             | 82  | 79  | 78  | 77 | 76 | 72 | 68 |
|                                     |                           | Outlet | 66                                        |                      |      |                                 | 74                             | 85  | 82  | 85  | 83 | 81 | 77 | 69 |
| 636                                 | 15                        | Inlet  | 50                                        | -                    | -    | 1.10                            | 65                             | 73  | 69  | 67  | 66 | 65 | 61 | 57 |
|                                     |                           | Outlet | 56                                        |                      |      |                                 | 68                             | 76  | 72  | 74  | 72 | 70 | 66 | 58 |
| 638                                 | 12                        | Inlet  | 45                                        | -                    | -    | -                               | 65                             | 67  | 65  | 61  | 61 | 59 | 55 | 49 |
|                                     |                           | Outlet | 51                                        |                      |      |                                 | 69                             | 71  | 68  | 70  | 67 | 64 | 59 | 50 |

\* Amperages for standard TE motors can be obtained at time of order. Check fan nameplate for exact amperage of all motors.

\*\* Test and Performance data are based on PowerLine Ultra

## POWERLINE ULTRA (PUD DOUBLE SKIN)

## Notes

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## Notes

[illegible]

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## Notes

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